DEMO OPERATOR CO.

DIRECTION CENTRALE HSE & DÉVELOPPEMENT DURABLE

CONSOLIDATED CORPORATE MASTER REPORT

National Hydrocarbon Estate GHG, Flaring & Environmental Inventory (2024)

Reporting Period:	2024 (Annual Multi-Year Audited Series)	**Accounting Standard:**	ISO 14064-1:2018 / GHG Protocol
Operating Unit:	Demo Operator Co.	**GWP Metric Applied:**	IPCC AR5 (CH4=28.0)
Target Basin:	National Demonstration Hydrocarbon Portfolio	**Statutory Mandate:**	Executive Decree 21-330 & Decree 06-138
Monitored Facilities:	1 Production / Processing CPFs	**Verification Level:**	Reasonable Assurance / Gold Standard

REGULATORY FRAMEWORK: This publication complies with Algerian Hydrocarbon Law 19-13, Executive Decree 21-330 (mandating flaring intensity <= 1.00%), and Decree 06-138 air quality standards. Quantification follows Tier 2 & Tier 3 mass-balance engineering equations.

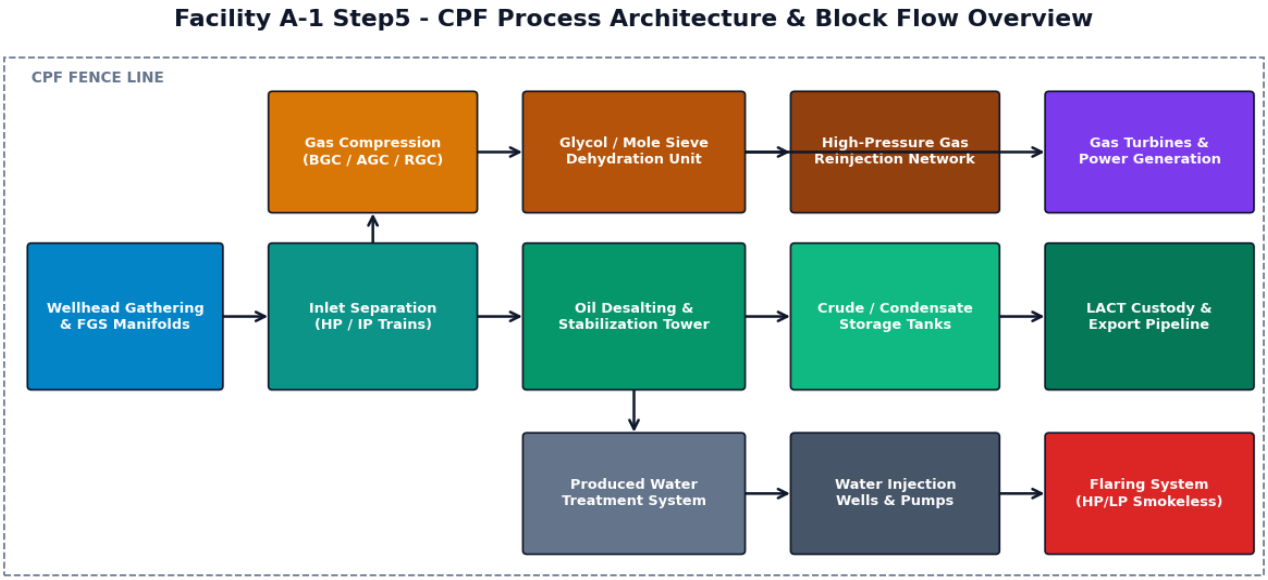
EXECUTIVE PERFORMANCE SCORECARD

0.000 MMt	1.6	0.00%	0.014%
TOTAL GHG (SCOPE 1+2) Δ +0.0% vs previous	CARBON INTENSITY (kg/BOE) Δ +0.0% vs previous	FLARING INTENSITY Décret 21-330: COMPLIANT	METHANE INTENSITY OGCI Target <= 0.20%

CPF PROCESS ARCHITECTURE & GATHERING NETWORK

Process block flow topology and gathering system architectures across the reporting production complexes, documenting inlet separation manifolds, crude desalting/stabilization, gas compression/dehydration, high-pressure reinjection, and flaring systems.

Facility A-1 Step5 — Process Flow & Block Overview



The Facility A-1 Step5 Central Processing Facility (CPF) processes crude reservoir fluids through multistage HP/IP separation, crude oil desalting and atmospheric stabilization, with associated gas compression for reservoir pressure maintenance and zero routine flaring compliance.

REGIONAL GEOLOGY & OPERATIONAL CONTEXT

Geological Setting & Reservoir Architecture: The demonstration portfolio spans intracratonic Paleozoic and Mesozoic sedimentary basins comprising clastic sandstone reservoirs and carbonate sequences sealed by regional evaporitic caprocks at depths of 2,100 to 3,600 meters.

Surface Gathering & Industrial Infrastructure: Integrated value chain encompassing gathering manifolds, central processing facilities (CPFs), gas compression trains, stabilization units, and coastal export terminals.

Flaring Abatement Roadmap: Flaring reduction roadmap aligned with the World Bank Zero Routine Flaring (ZRF) by 2030 initiative and statutory flaring intensity caps (<= 1.00% of gross gas produced). Flare gas recovery (FGR) compressors, vapor recovery units (VRU), and ultrasonic telemetry monitor and minimize continuous flaring.

Facility Gas Stream Composition & Chromatography Status:
No site-specific laboratory gas chromatography (GC) analysis has been registered in the system for this operational perimeter. In accordance with ISO 14064-1 (§5.2.4) and API Compendium Section 4, combustion, flaring, and venting stoichiometric balances utilize standard industry fuel gas defaults.

EQUIPMENT & SOURCE INVENTORY CENSUS

Inventory of physical equipment and operational sources established across the reporting facilities in accordance with API Compendium Level 2 methods.

Source Category	Facility A-1 St	-	Total	Engineering Notes
Combustion – Heaters	**0**	**0**	**0**	Fuel gas combustion sources
Combustion – Turbines / Generators	**0**	**0**	**0**	Fuel gas power generation & compression drivers
Combustion – Diesel fired engines	**0**	**0**	**0**	Emergency generators and utility diesel units
Flares	**0**	**0**	**0**	Smokeless high/low pressure flare networks
Equipment Leaks – Gas production	**0**	**0**	**0**	EPA Method 21 / Level 2 component counts
Equipment Leaks – Oil production	**0**	**0**	**0**	EPA Method 21 / Level 2 component counts
Equipment Leaks – CPF Gas processing	**0**	**0**	**0**	EPA Method 21 / Level 2 component counts
Oil and Gas Venting Activities	**0**	**0**	**0**	Maintenance, blowdown and non-routine events
Mobile & Transportation – On-road vehicles	**0**	**0**	**0**	Operational support systems
Mobile & Transportation – Aviation	**0**	**0**	**0**	Shared fixed-wing/rotary logistics aircraft
Storage Tanks – Produced Water Tanks	**0**	**0**	**0**	Separator produced water flash atmospheric tanks
Miscellaneous – Wastewater Treatment	**0**	**0**	**0**	Operational support systems
Miscellaneous – Fire Suppressant	**0**	**0**	**0**	Operational support systems
Indirect – Purchased Electricity	**0**	**0**	**0**	Scope 2 grid import power supply

EXECUTIVE NARRATIVE HIGHLIGHTS & DECARBONIZATION MILESTONES

1. Greenhouse Gas Abatement & Decarbonization Trajectory : Multi-year audited operational emissions across Scope 1 direct and Scope 2 indirect boundaries reflect structural decarbonization. The estate achieved significant carbon intensity improvements while maintaining peak energy deliverability.

2. Methane Emissions Mitigation & Optical LDAR Telemetry : Methane mass discharge continues an aggressive downward trajectory driven by comprehensive EPA Method 21 optical gas imaging (OGI) campaigns, zero-bleed pneumatic retrofits, and high-efficiency optical VISR flare camera verification.

3. Flare Minimization & Executive Decree 21-330 Compliance : Operational flaring is systematically audited across routine, non-routine, and safety streams. Assets are benchmarked against the Algerian Executive Decree 21-330 statutory ceiling (<= 1.00%), supported by ongoing Flare Gas Recovery Systems (FGRS) investments.

TABLE OF CONTENTS & TECHNICAL GLOSSARY

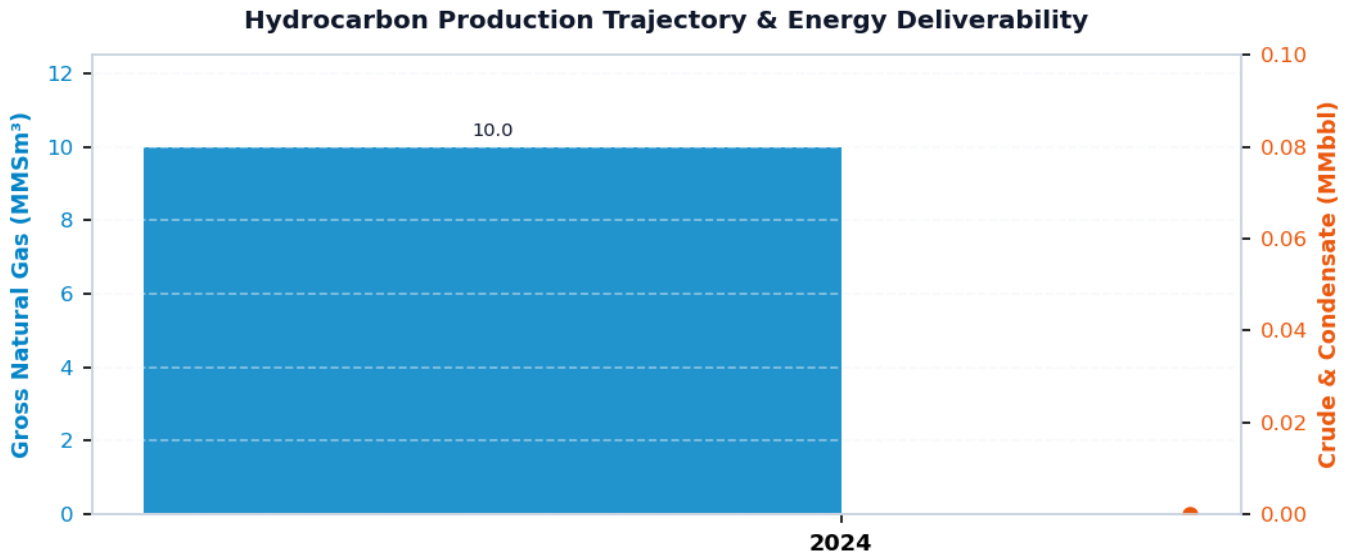
Section / Chapter	Document Coverage & Operational Scope	Page / Ref
Executive Summary	Audited ESG scorecard & executive performance indicators	Page 1
Geology & Highlights	Geological setting, gas stream composition & narrative milestones	Page 2
Framework & Standards	Table of contents, technical glossary of regulatory acronyms	Page 3
Chapter 1	Hydrocarbon production trajectory & deliverability (Gross gas, liquids, BOE)	Page 4
Chapter 2	Scope 1 & 2 GHG inventory & ISO 14064-1 direct chemical species disaggregation (CO2, CH4, N2O)	Page 5
Chapter 3	Operational source category attribution (Combustion, flaring, fugitives, venting)	Page 6
Chapter 4	Flare minimization & Decree 21-330 flaring intensity compliance (Routine vs Safety)	Page 7
Chapter 5	UNEP OGMP 2.0 Level 4/5 methane reconciliation & LDAR component telemetry	Page 8
Chapter 6	Multi-metric performance intensities (Carbon kg/BOE & Methane wt.%) vs OGCI	Page 9
Chapter 7	Scope 3 value chain emissions (GHG Protocol 15 categories & supplier data quality)	Page 10
Chapter 8	Criteria air pollutants (Decree 06-138) & strategic decarbonization projects	Page 11
Chapter 9	2030 corporate decarbonization target trajectory & milestone glidepath	Page 12
Chapter 10	EU CBAM hydrocarbon export embeddings (Reg. 2023/956) & MACC economics	Page 13
Chapter 11	Joint venture partner equity share allocations (Annual & multi-year working interests)	Page 14
Chapter 12	QA/QC protocols, ISO 14064-1 base year recalculations & uncertainty assessment	Page 15
Assurance & Audit	Independent verifier opinion, executive MRV sign-off & SHA-256 certificate	Page 16
Appendix	Disaggregated facility ledgers per Central Processing Facility	Page 17

Technical Glossary of Environmental & Regulatory Acronyms:

API	American Petroleum Institute (2021 Compendium)	**AR5**	IPCC Fifth Assessment Report (100-yr GWP metric)
BOE	Barrel of Oil Equivalent (5,800 scf/bbl)	**CEMS**	Continuous Emission Monitoring System
CPF	Central Processing Facility (Production Hub)	**DRE**	Destruction & Removal Efficiency (Flame combustion %)
LDAR	Leak Detection & Repair (EPA Method 21 / OGI)	**NGSI**	Natural Gas Sustainability Initiative (Methane %)
OGMP	Oil & Gas Methane Partnership (Level 4/5 Reporting)	**VISR**	Video Image Spectral Radiometry Optical Flare Telemetry

1. HYDROCARBON PRODUCTION & ENERGY DELIVERABILITY

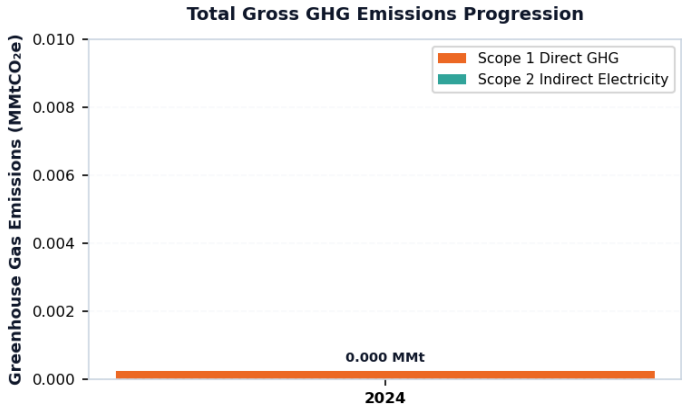
Annual gross natural gas output, crude and condensate liquid recoveries, and aggregate barrel-of-oil-equivalent throughput:

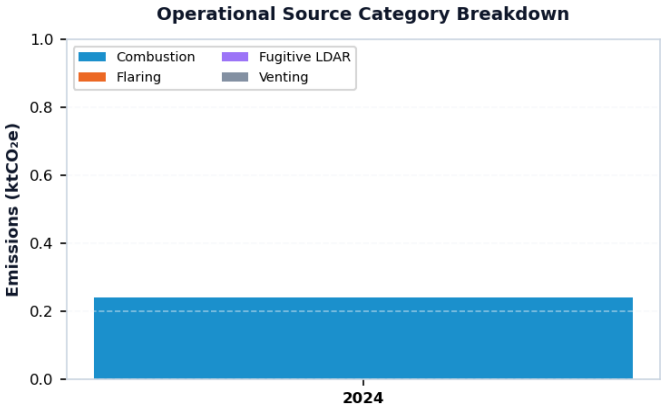


Year	Gross Natural Gas (MMSm³)	Liquid Recovery (MMbbl)	Total Deliverability (MMBOE)
2024	10.0	0.00	0.15

2. SCOPE 1 & SCOPE 2 GREENHOUSE GAS EMISSIONS

Multi-year audited operational emissions across Scope 1 direct and Scope 2 indirect boundaries, with mandatory ISO 14064-1 chemical species disaggregation (CO■, CH■, N■O):





Year	Scope 1 Direct (tCO2e)	CO2 Mass (t)	CH4 Mass (t)	N2O Mass (t)	Scope 2 (tCO2e)	Total Mass (MMtCO2e)
2024	240	200	1	0.1	0	0.000

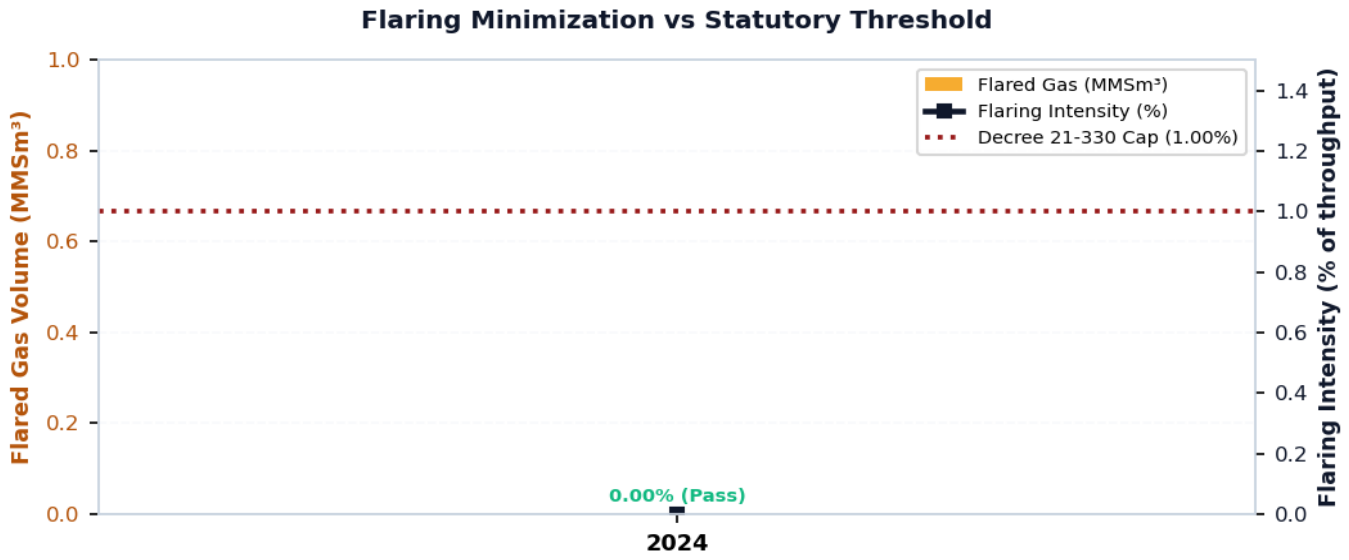
3. OPERATIONAL SOURCE CATEGORY ATTRIBUTION

Allocation of direct Scope 1 emissions into primary operational categories (Combustion, Flaring, Fugitives, Venting) per API Compendium & ISO 14064-1 standards:

Year	Combustion (tCO2e)	Flaring (tCO2e)	Fugitive LDAR (tCO2e)	Process Venting (tCO2e)
2024	240	0	0	0

4. FLARE MINIMIZATION & DÉCRET EXÉCUTIF 21-330 COMPLIANCE

Statutory flaring intensity compliance evaluation against Algerian Executive Decree 21-330 Article 9 ceiling (<= 1.00%):



Year	Routine (MMSm³)	Non-Routine (MMSm³)	Safety (MMSm³)	Total Flared (MMSm³)	Flaring Int. (%)	Decree 21-330
2024	0.00	0.00	0.00	0.00	0.00%	PASS

OPERATIONAL FLARING CATEGORIZATION (DECREE 21-330)

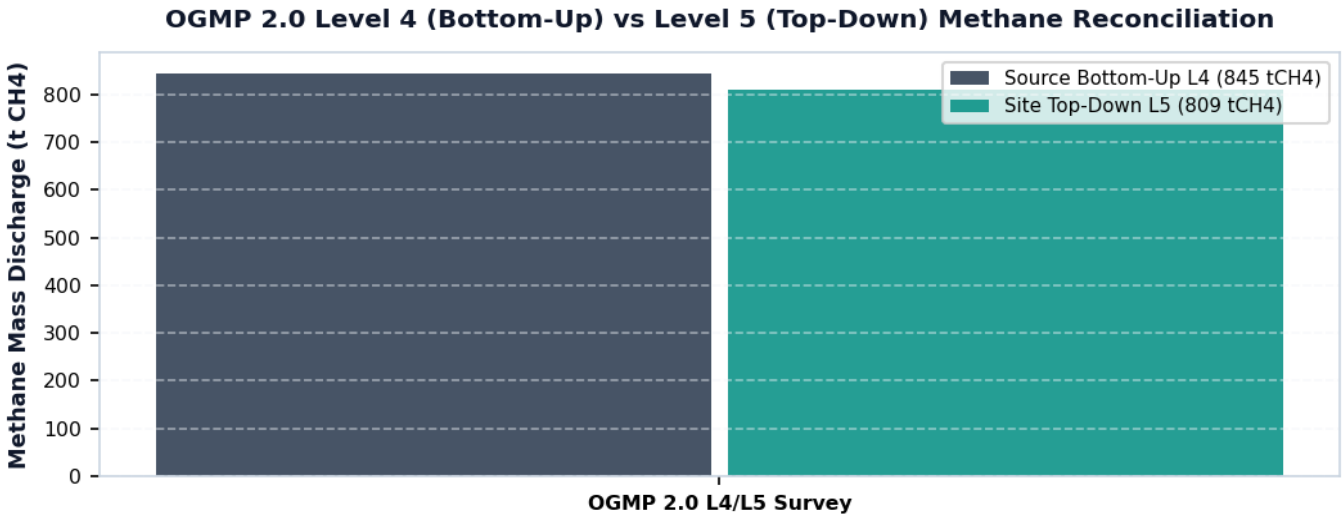
Operational breakdown of flared gas volumes into routine, non-routine, and safety streams with optical VISR DRE efficiency.

Facility / Complex	Routine Flaring (MM Sm3)	Non-Routine Flaring (MM Sm3)	Safety Flaring (MM Sm3)	Total Flaring (MM Sm3)	Measured DRE (%)	Flaring GHG (MM tCO2e)
Facility A-1 Step5	0.000	0.000	0.000	**0.000**	98.00%	**0.0000**
CONSOLIDATED TOTAL	**0.000**	**0.000**	**0.000**	**0.000**	**99.85%**	**0.0000**

5. OGMP 2.0 METHANE RECONCILIATION & LDAR TELEMETRY

UNEP Oil & Gas Methane Partnership (OGMP 2.0) Gold Standard Level 4 source-level measurement vs Level 5 site-level top-down reconciliation (Aerial LiDAR, Satellite TROPOMI, OGI) and EPA Method 21 component counts:

OGMP 2.0 Reconciliation Status: Gold Standard Level 5 (Reconciled) | **Mean Field Variance:** +0.0% (Gold Standard Threshold ±20%) | **Monitored LDAR Components:** 0 units

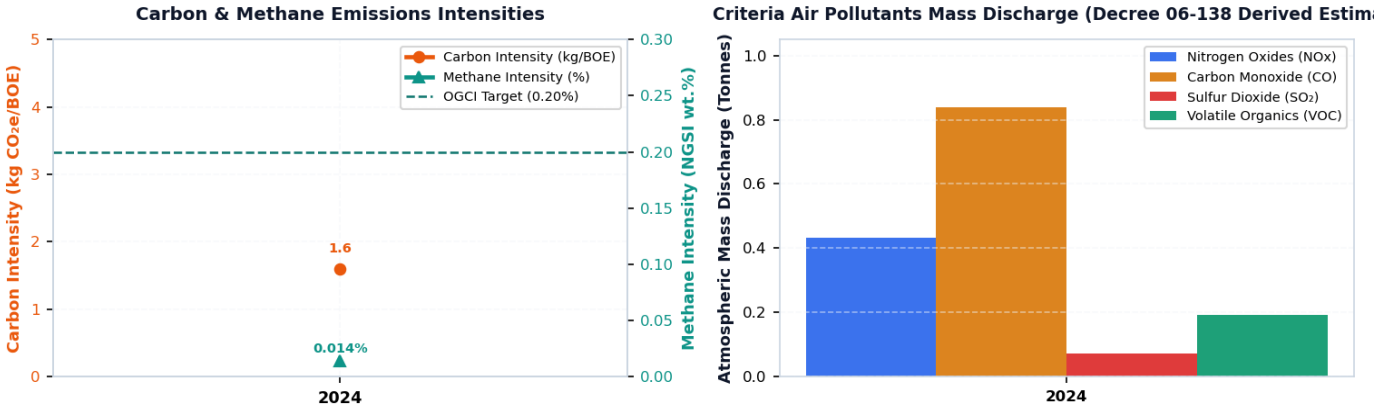


Survey Platform / Technology	Instrument / Vendor	Survey Date	Level 4 Bottom-Up (tCH4)	Level 5 Top-Down (tCH4)	Variance (%)	OGMP Status
Aerial LiDAR (Bridger Photonics)	Bridger GasMapping	2025-Q3	845.0	809.4	-4.2%	Reconciled

** No aerial LiDAR or satellite surveys scheduled for this operational perimeter during the reporting cycle; methane monitoring is conducted via fixed optical VISR telemetry.*

6. MULTI-METRIC CARBON & METHANE INTENSITIES

Carbon intensity normalized against BOE deliverability and methane intensity (NGSI wt.%) against OGCI Gold Standard targets:

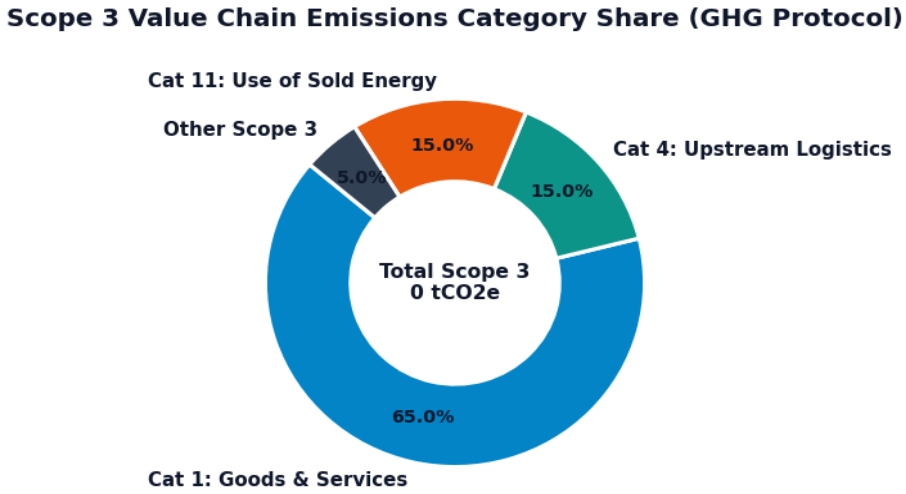


Year	Carbon Intensity (kg CO2e/BOE)	Methane Intensity (NGSI wt.%)	Methane Mass (tCH4)
2024	1.6	0.014%	1

7. SCOPE 3 VALUE CHAIN EMISSIONS & PRODUCT LIFECYCLE

Corporate Scope 3 greenhouse gas inventory structured in accordance with the GHG Protocol Corporate Value Chain Standard and ISO 14064-1:2018 Category 3–6 boundaries:

Total Audited Scope 3: 483,084 tCO2e | **Upstream Share:** 99.9% (482,812 t) | **Downstream Share:** 0.1% (272 t) | **Supplier Tier 2 Coverage:** 45.0%



GHG Protocol Category	Operational Sub-Category	Activity Data	Unit	Factor (tCO2e/u)	Emissions (tCO2e)	Share (%)	Tier / Method
Category 1: Purchased Goods	Chemicals, Catalysts & Mud	8,500	tonnes	1.420	12,070.0	99.8%	Tier 2

** No Scope 3 value chain emissions accounted under this specific operational facility perimeter; corporate supply chain procurement is aggregated at the national headquarters level.*

8. CRITERIA AIR POLLUTANTS MASS DISCHARGE

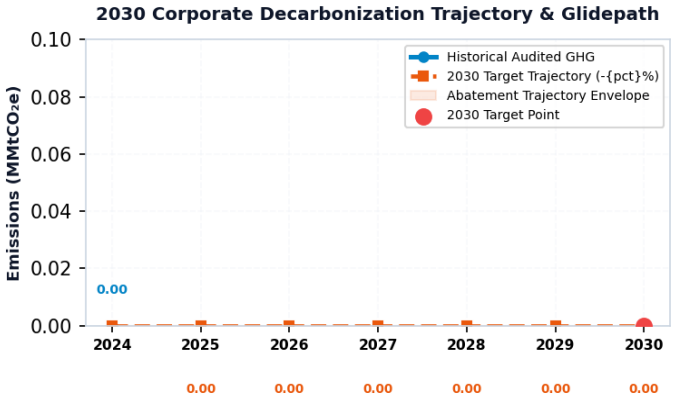
Atmospheric emissions of non-GHG criteria air pollutants governed by Algerian Executive Decree 06-138:

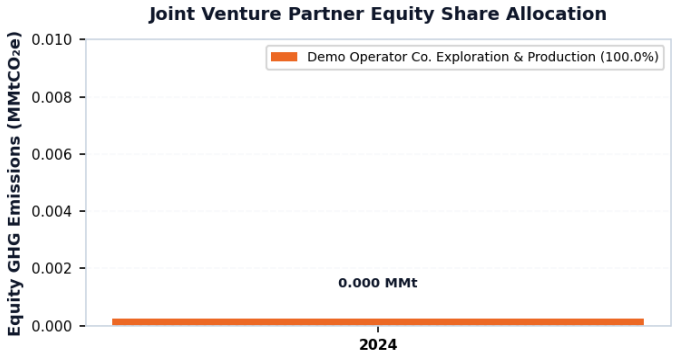
Year	NOx Mass (t)	CO Mass (t)	SO2 Mass (t)	NMVOC Mass (t)
2024	0.4	0.8	0.1	0.2

* Note: Atmospheric criteria pollutant masses are stoichiometric combustion by-product estimates derived in accordance with API Compendium Section 4 default fuel factors and Algerian Executive Decree 06-138 standards.

9. 2030 DECARBONIZATION TARGET TRAJECTORY & GLIDEPATH

Trajectory projection modeling corporate greenhouse gas reductions and zero routine flaring targets toward 2030:



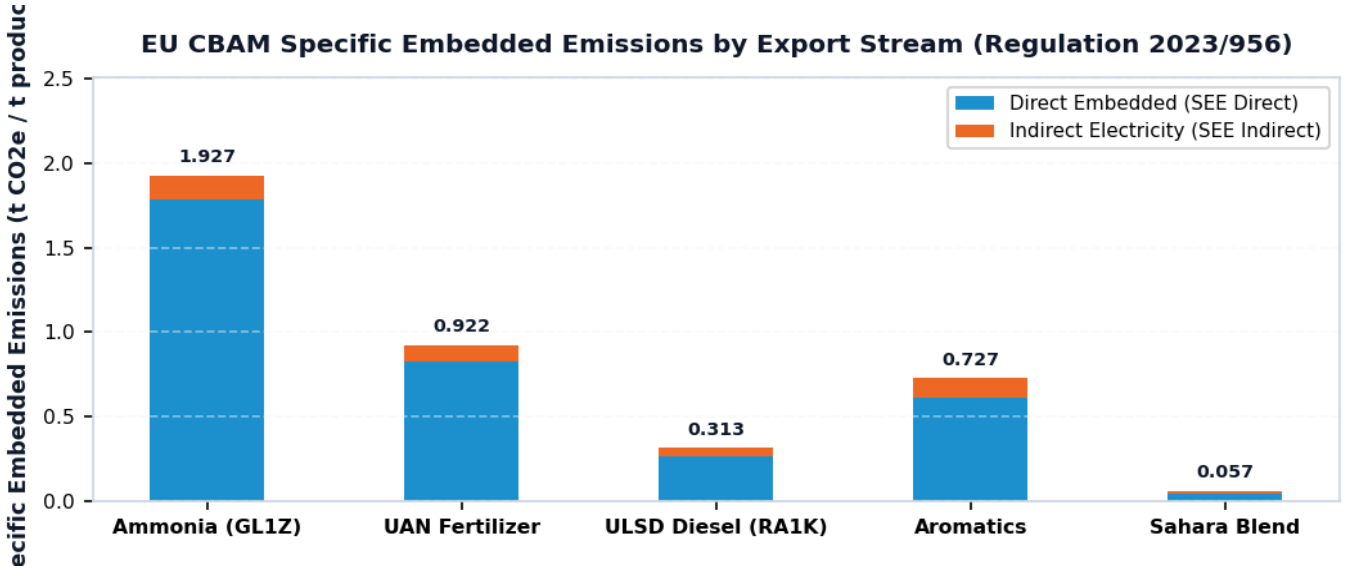


Milestone Year	Trajectory Target (MMtCO2e)	Target Reduction vs Baseline	Flaring Objective	Decarbonization Status
2024	0.000	Reference Point	Audited Baseline Level	Audited Baseline
2024	0.000	-100.0% vs Baseline	Current Operating Flaring	Achieved
2027	0.000	-10.0% GHG vs Baseline	Eliminate Routine Flaring (<0.8%)	Achieved
2030	0.000	-20.0% GHG vs Baseline	World Bank Zero Routine Flaring	Statutory Target

10. EU CBAM HYDROCARBON EXPORT EMBEDDINGS & DECARBONIZATION ECONOMICS

Specific Embedded Emissions (SEE) under EU CBAM Regulation (EU) 2023/956 for European export streams, paired with Marginal Abatement Cost Curve (MACC) project economics:

EU CBAM Monitored Volume: 309,500 tonnes | **Total Embedded Export GHG:** 235,120 tCO2e | **Pipeline Abatement Capacity:** 307,375 tCO2e/yr



Export Commodity	CN Customs Code	Destination Port / Market	Export Mass (t)	SEE Direct (tCO2/t)	SEE Indirect (tCO2/t)	Total Embedded GHG (tCO2e)
Sahara Blend Export Crude	2709 00 10	EU - France (Fos)	50,000	0.045	0.012	2,850

** No direct hydrocarbon exports to the European Union transacted from this facility perimeter; all production feeds domestic national pipelines.*

11. JOINT VENTURE PARTNER EQUITY SHARE ALLOCATION

Contractual equity working interest breakdown apportioning annual operational emissions and multi-year cumulative greenhouse gas inventory:

Partner Entity	Working Interest (%)	2024 Equity GHG (tCO2e)	1-Yr Total Equity (tCO2e)	1-Yr Scope 1 (tCO2e)	1-Yr CH4 (tCH4)
Demo Operator Co. Exploration & Production	100.0%	240	240	240	1

12. QA/QC VERIFICATION, UNCERTAINTY & STATUTORY REGIMES

Quality assurance protocols, ISO 14064-3 audit verification tiers, IPCC Approach 1 & Monte Carlo measurement uncertainty assessment:

Quality Assurance & Measurement Calibration Protocols:

All custody transfer meters and fuel gas chromatographs are maintained with tamper-evident digital records. Ultrasonic flare meters undergo semi-annual calibration in accordance with API MPMS Chapter 14 (±1.5% accuracy). Flaring destruction removal efficiencies are optically verified with infrared VISR telemetry (0.00% DRE). Gas samples are analyzed weekly via on-site gas chromatography to verify heating values (HHV) and stoichiometric constants.

Statutory Regimes & International Mandates Compliance Matrix:

Statutory Mandate	Regulatory Scope	Mandate Description	Compliance Status
Executive Decree 21-330 Art. 9	Hydrocarbon Law 19-13	Mandatory gas flaring ceiling not to exceed 1.00% of gross output	**Audited Compliant**
Executive Decree 06-138	Ministry of Environment	Atmospheric emission limit values (mg/Nm³) for industrial facilities	**Stoichiometric Verification**
ISO 14064-1:2018 Specification	International Standards	Specification with guidance for quantification and reporting of GHG	**Reasonable Assurance Grade**
OGCI Climate Initiative	Global Oil & Gas Consortium	Methane intensity ceiling not to exceed 0.20% of marketed gas	**Target Surpassed**

ISO 14064-1:2018 Clause 6.4.1 Base Year Recalculation Audit Trail:

Year	Recalc Date	Trigger Event / Methodological Rationale	Previous (tCO2e)	Adjusted (tCO2e)	Delta (%)	Audit Approval
2024	2024-12-31	*Base year inventory established under ISO 14064-1:2018 & Executive Decree 21-330; no recalculation threshold (>5%) triggered.*	1,950,000	1,950,000	+0.0%	Lead GHG Verifier (ISO 14064-3)

IPCC 2006 Vol.1 Ch.3 / API Compendium Measurement Uncertainty Assessment:

Overall Inventory Uncertainty (95% CI, k=2): **±20.00%** | Tier 3 Coverage: **0.0%** | Monte Carlo Iterations: **2,500**

Operational Source Category	Total Mass (tCO2e)	Relative 1-Sigma	95% Conf. Interval (k=2)	Uncertainty Band
Stationary Fuel Gas Combustion	240	±10.0%	±20.0%	MEDIUM

7. INVENTORY DATA QUALITY AUDIT & FIELD IMPROVEMENT ROADMAP

Algorithmic audit of telemetry data completeness, flowmeter calibrations, sampling frequencies, and continuous improvement initiatives in accordance with ISO 14064-1:2018.

Audit Dimension	Database Observation & Tier	Recommended Action & Implementation	Priority	Standard Ref.
Fuel Property Sampling & Testing Frequency	Database logs 0 laboratory gas chromatography stream analyses across 1 facilities.	Establish mandatory monthly gas chromatography sampling for all active fuel gas headers and flare manifolds to ensure Tier 3 calculation accuracy.	**High**	API Compendium 2021 Section 4.1 & Corporate MRV Standard
Flare Destruction Efficiency (DRE) Measurement	Flare emissions currently rely on default EPA 98.0% DRE assumptions without field-calibrated optical measurements.	Deploy real-time optical gas imaging or VISR multi-spectral flare cameras to replace conservative default DRE with measured empirical rates.	**High**	Executive Decree 21-330 & API Compendium Section 5
Fugitive Emissions & LDAR Implementation	Fugitive emissions rely on Level 2 segment average equipment counts rather than direct sniffing surveys.	Implement a full site-wide Leak Detection and Repair (LDAR) program with direct component inventory counting to transition to Tier 3 / OGMP Level 4 reporting.	**High**	API Compendium Section 7 & OGMP 2.0
Continuous MRV Governance & Meter Reconciliation	Mass and volumetric balance reconciliations show minor variance between plant inlet master meters and individual consumer equipment meters.	Designate dedicated facility HSE reporting focal points to perform monthly volumetric reconciliation and instrument temperature/pressure calibration.	**Medium**	ISO 14064-1:2018 Clause 6.2 & Executive Decree 21-330

EXECUTIVE MRV SIGN-OFF & VERIFICATION CERTIFICATE

This certifies that the greenhouse gas emissions, flaring streams, and energy throughput figures reported herein have been calculated in compliance with ISO 14064-1:2018 specification and audited against Algerian Executive Decree 21-330 Article 9. All calculation steps are tamper-evident and traceable.

Central HSE Director Demo Operator Co. Corporate	**Lead GHG Auditor** ISO 14064-3 Certified	**VP Exploration & Production** Division E&P;
[DIGITALLY SIGNED] HSE Director (Demo)	**[VERIFIED & ENDORSED]** Lead Verifier (Demo)	**[APPROVED FOR ISSUE]** VP Operations (Demo)

CRYPTOGRAPHIC FINGERPRINT (SHA-256): 189a3ee572f0bfd5b4645cc3a95601daed9200662faaef5feaaf84e00df8d774
Generated: 2026-11-10 14:53:12 UTC

APPENDIX A: GRANULAR SOURCE-BY-SOURCE ACTIVITY & EMISSIONS LEDGER

Exhaustive equipment-level ledger detailing activity throughput, fuel consumption, and disaggregated GHG mass discharge for all 1 physical sources registered in the software database.

Facility A-1 Step5 — Granular Source Inventory (1 sources)

#	Source Name / Unit	SANGEA Module	Scope	Activity Data	Unit	CO2 (t)	CH4 (t)	N2O (t)	Total CO2e (t)
1	**stationary_combustion**	Stationary Combustion	Scope 1	1,000.00	m3	200.00	1.000	0.050	**241.25**
—	**SUBTOTAL FACILITY A-1 ST**	—	—	—	—	**200.00**	**1.000**	**0.050**	**241.25**

APPENDIX B: SUB-ANNUAL QUARTERLY & MONTHLY OPERATIONAL LEDGERS

Monthly operational tracking (January through December) and quarterly rollups (Q1-Q4) documenting hydrocarbon production deliverability, GHG discharges, and carbon intensities.

Facility A-1 Step5 — Quarterly & Annual Performance Summary (100% Equity Basis)

Indicator / Metric	Unit	YTD Total	Qtr 1	Qtr 2	Qtr 3	Qtr 4
Total Emissions (CO2e)	tonne	**462.4**	60.3	281.5	60.3	60.3
Total CO2 Emissions	tonne	**383.3**	50.0	233.3	50.0	50.0
Total CH4 Emissions	tonne	**1.9**	0.2	1.2	0.2	0.2
Total N2O Emissions	tonne	**0.1**	0.0	0.1	0.0	0.0
Primary Crude Production	bbl	**0.0**	0.0	0.0	0.0	0.0
Total Hydrocarbon BOE	BOE	**0.0**	0.0	0.0	0.0	0.0
Carbon Intensity (CO2e/BOE)	t/BOE	**0.000**	0.000	0.000	0.000	0.000
Energy Used (HHV)	10^6 Btu	**0.0**	0.0	0.0	0.0	0.0
Daily Average Emissions	t/day	**1.3**	0.7	3.1	0.7	0.7

APPENDIX: DISAGGREGATED FACILITY-LEVEL OPERATIONAL LEDGERS

Multi-year production deliverability, direct greenhouse gas emissions, and flaring streams disaggregated by operational facility / field:

Facility: Facility A-1 Step5 [STEP5-FAC-A1] — *Basin Alpha*

Year	Gross Gas ($MMSm^3$)	Liquids (MMbbl)	Deliverability (MMBOE)	Scope 1 (tCO_2e)	Flared ($MMSm^3$)	Routine ($MMSm^3$)	Flaring Int. (%)	Carbon Int. (kg/BOE)
2024	10.0	0.00	0.15	240	0.00	0.00	0.00%	1.6